When Solar Projects Fail: Belief Reversal and Farmland Prices

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Abstract

When proposed solar projects enter an interconnection queue, they become publicly observable and capitalize an energy development option into surrounding farmland prices. Project withdrawal reverses this expectation. Using approximately 68,000 parcel-level farmland transactions from Indiana and Illinois from 2016 to 2022, we estimate the causal effect of this belief-reversal mechanism on farmland prices. Because project withdrawals are endogenous to local land market conditions, we instrument failed solar capacity using network upgrade costs assigned during the interconnection study process. Our IV estimates show that failed solar capacity reduces farmland prices by approximately 6.8 percent at the sample mean, or roughly $1,389 per acre, a magnitude consistent with partial capitalization of the energy development option. This average effect masks substantial heterogeneity across local land markets. In nonmetro and low-population counties, project failure generates large price declines, while in metro and more urbanized counties, the effect is near zero, as the loss of the energy option is offset by restored access to alternative land uses. These findings provide the first causal evidence of belief reversal in agricultural land markets and highlight that the welfare costs of solar project failure fall disproportionately on rural communities.

Keywords: Farmland prices; Solar energy development; Belief reversal; Instrumental variables; Rural–urban heterogeneity

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1 Introduction

Asset prices reflect not only current fundamentals, but also expectations about future economic conditions (Fama, 1970; Campbell, 1991). Because expectations are formed under uncertainty, they are continuously updated as new information arrives (Lucas Jr, 1972; Sargent et al., 1973). As a result, asset prices adjust not only when outcomes are realized but also when beliefs about those outcomes change. If the anticipated events do not materialize, agents revise their prior expectations downward, generating negative information shocks that are reflected in price. This price effect is likely to be especially important in thin, spatially segmented markets like farmland, where transactions are infrequent, price discovery is slow, and prices depend heavily on expectations about future land use (Just and Miranowski, 1993; Plantinga et al., 2002).

The rapid expansion of utility-scale solar energy across the Midwest provides an unusually clean setting in which to study this mechanism. These projects introduce an alternative land use option, under which farmland is used for energy production under long-term lease agreements, often over 25 to 35 years, and with lease rates three to five times the prevailing cropland cash rents (Maguire et al., 2024; Winikoff and Parker, 2024; Fiechter et al., 2024). This long-term financial premium shifts landowner expectations about future land use and is capitalized into farmland prices even before projects are completed (Abashidze and Taylor, 2023; Plantinga et al., 2002; Kunwar, 2024; Haan and Simmler, 2018). However, a large proportion of proposed projects that enter interconnection queues are never completed, with many withdrawn prior to construction due to permitting constraints, financing challenges, or transmission limitations (Rand et al., 2024). When these projects fail, long-horizon expectations embedded in the land price must be revised downward, potentially reducing the option value previously capitalized in land values. These dynamics highlight the role of expectations in determining land values and motivate the question of how farmland markets respond when expected development does not materialize.

A growing body of empirical literature examines the impact of renewable energy development on farmland prices and land use. Several of these studies document that the proximity to solar or wind energy is capitalized into farmland prices, consistent with expectations of higher returns from conversion or lease opportunities (Haan and Simmler, 2018; Abashidze and Taylor, 2023; Kunwar, 2024). Haan and Simmler (2018) exploit cross-sectional variation in wind potential interacted with Germany's feed-in tariff introduction and find that about 18% of expected wind turbine profits are reflected in agricultural land values, evidence that the option to host energy infrastructure is priced before any turbine is constructed. Lai et al. (2019) find that solar installations generate positive spillover effects on nearby farmland values, with price increases relative to average land prices, suggesting that completed solar development on one parcel raises the perceived energy development potential of neighboring land. Similarly, Abashidze and Taylor (2023) and Kunwar (2024) document that solar development raises surrounding farmland prices, even for parcels not directly hosting solar facilities. They attribute these effects to the increased option value available to landowners, as the possibility of future conversion to energy production is capitalized into current land values. Together, these findings confirm that land markets price expected future development opportunities, not only realized changes in land use. However, these studies do not answer the counterfactual: what happens to land prices when a proposed project enters the public record, raises local expectations, and is subsequently canceled before any construction occurs? If forward-looking expectations drive prices upward in anticipation of development, their reversal upon project failure should exert downward pressure, yet this mechanism remains entirely unexamined in the literature.

This paper studies how farmland prices respond when proposed solar projects are withdrawn,[1] and whether this effect varies across the rural-urban landscape. We focus on projects that enter interconnection queues, become publicly observable, and subsequently fail to materialize prior to construction. This failure reverses previously capitalized expectations regarding future land use. We exploit information on project withdrawals from the Midcontinent Independent System Operator (MISO) interconnection queue to construct measures of negative information shocks to farmland markets in Indiana and Illinois from 2016 to 2022. However, withdrawal decisions are endogenous to local conditions: developers select project locations based on unobserved site char-

[1] Prior research establishes that anticipated development capitalizes into farmland prices (Plantinga et al., 2002; Haan and Simmler, 2018; Abashidze and Taylor, 2023; Kunwar, 2024). We do not attempt to re-establish this here, nor do we estimate whether the observed price decline fully offsets prior appreciation. Both questions require instruments for the entry decision itself, which lies outside the scope of this analysis.

acteristics that may independently affect farmland prices. Similarly, projects are more likely to fail in locations with lower expected returns to development, weaker demand for land conversion, or regulatory and economic constraints that also directly affect farmland values. As a result, a direct comparison between areas with and without project failure would confound the causal effect of project withdrawals with pre-existing differences in local fundamentals. To address this endogeneity, we instrument for failed project capacity using the network upgrade costs assigned during the interconnection study process. These costs are primarily determined by engineering requirements for connecting a project to the existing transmission grid, such as line capacity, grid topology, and system load, and are, critically, unknown and difficult to predict prior to queue entry (Johnston et al., 2023). These cost realizations are plausibly orthogonal to local farmland market conditions, and projects assigned with higher network upgrade costs are significantly more likely to be withdrawn (Seel et al., 2022; Johnston et al., 2023), providing a credible first stage. This approach isolates the variation in project failure driven by grid-engineering constraints, enabling identification of the causal effect of failed solar development on farmland prices. Using this instrumental variables strategy in combination with transaction-level farmland sales data, we estimate the causal effect of failed solar capacity on land values and examine how these effects vary across urban and rural counties.

The major finding in this study is that project failure reduces farmland prices on average, but the welfare cost falls disproportionately on rural communities. Ordinary least squares estimates are close to zero, consistent with negative selection in developer site choice. Instrumenting failed solar capacity with network upgrade costs yields negative and precisely estimated effects: our preferred specification implies a price decline of approximately 6.8 percent at the sample mean of failed capacity, or roughly $1,389 per acre. This magnitude is consistent with a simple back-of-the-envelope calculation of the option value of solar development, suggesting partial capitalization of the energy development premium at the time of queue entry. However, this average effect masks substantial heterogeneity driven by outside options. In nonmetro counties, where energy development represents one of the few opportunities for returns above traditional agricultural income, each additional megawatt of failed capacity reduces farmland prices by approximately

4.1 percent. In metro counties, the loss of the energy option is offset by the restored value of alternative residential and commercial uses, leaving the total effect near zero. This rural-urban contrast is not discrete, interacting failed capacity with county population reveals a monotonically increasing gradient in the price response. These results highlight the role of expectations and outside options in shaping land values and imply that the welfare cost of solar project failure falls disproportionately on agricultural areas with limited alternative development opportunities.

We make three major contributions to the literature. First, we provide the first causal estimates of how asset prices respond to failed development expectations in agricultural land markets. The existing literature on land markets shows that forward-looking beliefs are embedded in the land prices, but direct large-sample evidence on the downward revision of those beliefs is not present. By studying project withdrawals rather than completions, we isolate the belief-reversal mechanism and quantify its price consequences. Second, we introduce an empirical framework that uses interconnection queue data to distinguish between anticipated and failed development, and exploits exogenous variation in network upgrade costs as an instrument in a land market context. To the best of our knowledge, this is the first application of interconnection cost variation as an instrument for project failure in land valuation research. Third, we document significant heterogeneity in the price response across the rural-urban landscape. The price effect of project failure varies continuously with the availability of alternative land use options, declining monotonically from large negative effects in low-population agricultural counties to near-zero or positive effects in urban counties. This heterogeneity has direct implications for how the welfare costs of the attrition of energy projects are distributed throughout the rural-urban landscape.

The remainder of the paper is organized as follows. Section 2 describes the institutional background and data sources. Section 3 develops the conceptual framework. Section 4 presents the empirical strategy and identification approach. Section 5 reports and discusses the main results. Section 6 concludes.

2 Institutional Background and Data

2.1 The Interconnection Process

Utility-scale renewable energy projects must obtain permission to connect to the bulk power transmission grid before construction can begin. This process is known as interconnection, and is administered by regional transmission organizations (RTOs) and independent system operators (ISOs). RTOs and ISOs coordinate the operation of high-voltage transmission networks and wholesale electricity markets across multi-state regions. The Midcontinent Independent System Operator (MISO) is responsible for the operation of the bulk transmission system and wholesale electricity across all or parts of fifteen U.S. states, including Indiana and Illinois.[2] Interconnection queues consist of projects that have applied to connect to the grid and are undergoing a series of engineering studies to determine required network upgrades and associated costs (Rand et al., 2024).

When a developer proposes a new generation project, it submits an interconnection request to MISO specifying key project characteristics, including location, technology type, proposed capacity, point of interconnection, and expected commercial operation date. After submission, the project enters the MISO Generator Interconnection Queue (GIQ), where it becomes publicly observable and is tracked through the development process. The interconnection queue serves as a comprehensive dataset of proposed, active and completed projects, while also recording projects that were ultimately withdrawn from the queue, along with their withdrawal dates (Rand et al., 2024).

Before a queued project begins construction, it must complete a series of engineering studies that evaluate the feasibility and cost of grid connection. Feasibility study provides an initial assessment of system impacts and upgrade costs. System impact study evaluates the effects of the project on grid reliability and power flows across the transmission network. The facilities study

[2]MISO's footprint includes all or portions of Arkansas, Illinois, Indiana, Iowa, Kentucky, Louisiana, Michigan, Minnesota, Mississippi, Missouri, Montana, North Dakota, South Dakota, Texas, and Wisconsin. MISO also coordinates reliability with the Canadian province of Manitoba. Some counties in Indiana and Illinois are served by PJM rather than MISO; we include PJM queue records for those counties in the final project dataset.

determines the specific infrastructure requirements, cost estimates, and implementation timeline for interconnection. Together, these studies identify the network upgrades required to complete the project, such as transmission line expansions or substation investments or other grid infrastructure investments that must be constructed before the project can safely connect to the system. The costs of these upgrades are assigned to the developer as a condition of interconnection (Rand et al., 2024; Johnston et al., 2023).

Interconnection costs comprise two main components: the direct cost of connecting a project to the grid, and the indirect cost of upgrading the transmission network to accommodate new generator without overloading existing infrastructure (Johnston et al., 2023). These costs can vary significantly across projects as a function of local grid conditions, and are determined through engineering study rather than by local economic conditions or farmland market fundamentals. Johnston et al. (2023) show that these costs are difficult to predict ex ante [3]. Additionally, costs do not systematically decrease across successive studies, suggesting that each stage reveals new information rather than refining prior expectations. The distribution of costs is also highly right-skewed, with a median near zero but a 90^{th} percentile of approximately \$0.41 million per megawatt (Johnston et al., 2023).

This cost unpredictability is the central feature that motivates our identification strategy. Johnston et al. (2023) provide direct evidence that projects assigned with high intersection cost are significantly more likely to be withdrawn: projects with second-study costs exceeding 0.1 million per megawatt are about 49 percentage points more likely to exit the queue before completing the interconnection process, even after controlling for permitting difficulty, distance to the grid, and substation fixed effects. Because these cost realizations are driven by the physical characteristics of the transmission network rather than local land market conditions, they provide a plausibly exogenous source of variation in project failure, forming the basis of our instrumental variable (IV) strategy.

[3] In Johnston et al. (2023), a regression of second-study costs on observable project characteristics,including size, fuel type, location, and entry year, yields an R^2 of only 0.41, indicating that a large share of cost variation is unexplained by observable attributes. Although Johnston et al. (2023) focus on PJM, the underlying mechanisms they document, including cost uncertainty arising from engineering constraints and the effect of high upgrade costs on project withdrawal, are common features of interconnection processes across ISO/RTO operators, including MISO.

The scale of the interconnection queue has grown dramatically in recent years, which reflects the surge in renewable energy investment. By the end of 2023, approximately 2,600 gigawatts of total generation and storage capacity were actively seeking grid connection across the United States, with solar, wind, and storage accounting for nearly 95 percent of proposed capacity (Rand et al., 2024). This rapid growth has put significant pressure on the interconnection process, resulting in longer study timelines, increased network upgrade costs being assigned to projects, and historically high withdrawal rates. The typical duration from an interconnection request to commercial operation has increased from less than two years for projects built in the early 2000s to more than four years for projects built between 2018 and 2023 (Rand et al., 2024). Despite the large volume of proposed capacity, only a few projects are ultimately built. For example, only about 20 percent of the proposed projects that entered interconnection queues between 2000 and 2018 reached commercial operations (Rand et al., 2024). For local land markets, this means that a substantial portion of anticipated development never materializes. As a result, the interconnection queue provides a valuable setting for studying how farmland prices respond when expected development fails to occur.

2.2 Farmland Transaction Data

We use transaction-level farmland sales data to measure land prices and construct our main outcome variable. Our farmland price data are obtained from Acres, a proprietary agricultural land marketplace and data platform that compiles parcel-level records of farmland transactions across the United States.[4] This dataset includes Indiana and Illinois from 2016 to 2022 and records, for each transaction, the sale price per acre, sale date, parcel acreage, soil quality as measured by the National Commodity Crop Productivity Index (NCCPI), and spatial variables like distances to key infrastructure including transmission lines, substations, solar farms, and wind turbines. The NCCPI is a composite productivity index ranging from 0 to 100 that measures the inherent productivity of a soil profile for non-irrigated commodity crops, primarily corn, soybeans, and small

[4] Acres is an agricultural land data company that aggregates transaction records from county deed registries, assessor offices, and multiple listing services across the United States.

grains, and is a standard parcel-level control for land quality in the farmland valuation literature (Ifft et al., 2019; Dinterman and Katchova, 2019; Delbecq et al., 2014). We restrict the sample to arm's-length transactions by excluding same-name buyer-seller pairs, which likely reflect intra-family transfers rather than market-price sales, and remove observations with missing values on key covariates. The final estimation sample contains approximately 68,000 transactions across 92 counties in Indiana and 102 counties in Illinois.

2.3 Solar Project Data

Data on solar project development are obtained from the Generator Interconnection Queue (GIQ) maintained by the Midcontinent Independent System Operator (MISO). The MISO GIQ data are publicly available through MISO's website, and includes detailed project-level information, such as location (county and state), fuel type, proposed capacity, queue entry date, study status, interconnection cost estimates, and key milestone dates, including withdrawal and completion. The data also record the estimated network upgrade costs assigned to each project during the engineering study process, reported separately as point-of-interconnection costs and network upgrade costs, both in dollars per kilowatt. We obtain the full historical queue records for projects located in Indiana and Illinois covering our sample period of 2016 to 2022.

Two features of the MISO queue data warrant explicit acknowledgment. First, project locations are recorded at the county level rather than the parcel level. This is consistent with the interpretation of interconnection queue filings as county-level public information events, and motivates the construction of county-year treatment variables that are then merged with parcel-level transaction data for empirical analysis. Second, the analysis relies on administrative interconnection queue records, which provide information on project status and timing but do not measure the expectations or beliefs of individual market participants. The interpretation of project withdrawal as a negative expectation shock is therefore inferred from observed price responses rather than directly observed. This is an inherent limitation of using transaction data to study expectation formation, and is common to the broader literature on asset price responses to information events.

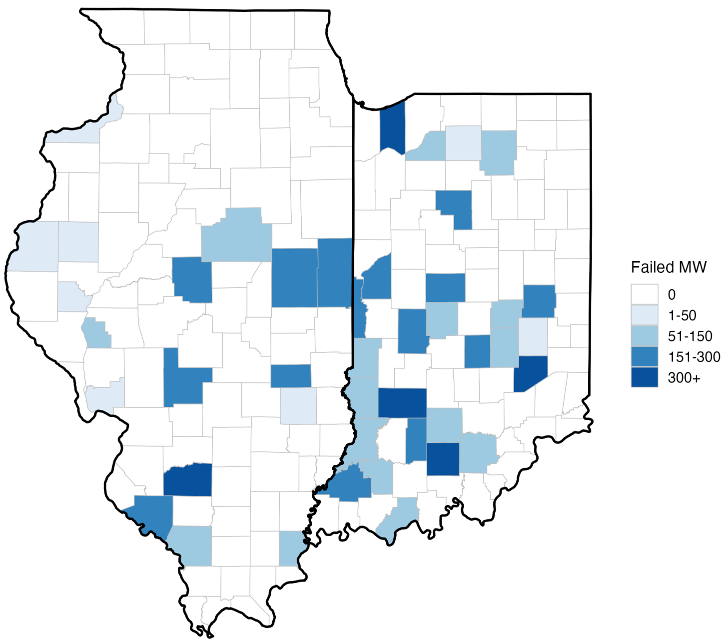
We merge the queue data with individual farmland transaction data using county FIPS codes. We construct three variables that measure the cumulative stock of solar capacity at different stages of the project lifecycle. *Queued MW* captures capacity that has entered the queue and remains neither completed nor withdrawn as of the transaction date, representing active but unresolved development expectations in the county. *Completed MW* measures the cumulative in-service capacity of projects that have reached commercial operations. *Failed MW*, our primary variable of interest, is the cumulative capacity of projects that have been withdrawn from the queue by the time of each farmland transaction, representing the stock of disappointed development expectations in the county. We use a cumulative stock measure because past withdrawals remain informative about the perceived viability of future solar development in the county. [5]

Figure 1 maps the geographic distribution of Failed MW across counties in Indiana and Illinois, illustrating that treatment variation is concentrated in a subset of counties and is not systematically clustered in any particular region of the two states. Figure A1 in the appendix documents the spatial distribution of solar projects by development status. Similarly, Figure A2 plots the timeline of solar project activity in the sample, showing the rapid growth in queue entries from 2016 onward and the corresponding rise in withdrawals in later years.

For the instrumental variable, we construct the average network upgrade cost weighed in MW (in $ / kW) for all projects active in the interconnection queue in county c at time t. This variable captures the engineering cost burden faced by the projects in a county's local grid zone. It varies across counties due to differences in transmission capacity and grid topology, and over time as system load and network congestion evolve with the expansion of renewable generation. Importantly, this variation is driven by the physical engineering requirements of the existing transmission system rather than by local economic or agricultural conditions. Table 1 reports summary statistics for the key variables used in the analysis.

[5] In the real-options framework, the value of the energy development option depends on the perceived probability that future development can be successfully realized. Cumulative failed capacity captures the accumulated negative signal about county-level solar viability that is capitalized into land prices. As a robustness check, we estimate specifications using annual withdrawal flows and continue to find evidence that project failure reduces farmland prices (Table A1).

Figure 1: Failed Solar Capacity by County



Notes: This figure shows the geographic distribution of the solar project capacity withdrawn (MW) from the MISO interconnection queue over the sample period. Darker shading indicates higher Failed MW; counties with no solar project withdrawals are shown in the lightest shade. Failed MW is defined as the total capacity of solar projects withdrawn from the MISO Generator Interconnection Queue in a county through the end of the sample period.

3 Conceptual Framework

3.1 Farmland Value and the Energy Development Option

According to the standard capitalization model, farmland value reflects the discounted value of expected future returns to its highest and best use (Melichar, 1979; Just and Miranowski, 1993). In a purely agricultural setting, this corresponds to the capitalized value of expected farm income, which is determined by agricultural fundamentals such as soil quality, crop prices, and input costs. The development of renewable energy on farmland introduces a competing land use:

Table 1: Summary Statistics

Variable	Mean	SD	Min	Max
Panel A: Farmland Transactions				
Log(Price/Acre)	8.555	1.302	4.52	17.381
Parcel Size (Acres)	81.8	223.7	0.04	32,702.7
NCCPI	64.8	16.7	0.006	97.4
Distance to Substation (miles)	2.53	1.04	0.002	9.33
Distance to Solar Farm (miles)	14.46	8.74	0.002	65.33
Transmission Line (miles)	1.93	2.19	0.001	17.29
Panel B: County-Year Solar Development				
Failed MW	17.23	66.27	0	544.1
Realized MW	6.94	39.48	0	475.0
Expected MW	72.21	172.39	0	1,748.0
Network Upgrade Cost ($/kW)	3.36	24.99	0	285.1

Notes: This table reports summary statistics for the estimation sample. Panel A reports parcel-level transaction statistics (N = 68,058). Panel B reports county-year level statistics for solar development variables (N = 1,018); the minimum of zero reflects that some county-years have no solar project activity. NCCPI is the National Commodity Crop Productivity Index (0–100), measuring soil productivity for non-irrigated commodity crops. Network Upgrade Cost is the average interconnection cost ($/kW) obtained from the Lawrence Berkeley National Laboratory interconnection cost dataset (Seel et al., 2022).

farmland close to the grid infrastructure can be converted to energy production under long-term lease agreements that offer returns much higher than the prevailing agricultural rents (Maguire et al., 2024; Fiechter et al., 2024). This alternative land use generates an option value in the sense of Dixit and Pindyck (1994), as conversion is irreversible, outcomes are uncertain, and timing is endogenous.

In this study, we model farmland prices reflecting a set of mutually exclusive land use outcomes. At time t, the price of parcel i can be written as:

$$P_i = V_i^{ag} + \underbrace{\pi_i^{energy}\left(V_i^{energy} - V_i^{ag}\right)}_{\text{energy option value}} + \underbrace{\pi_i^{alt}\max\left(V_i^{alt} - V_i^{ag},\ 0\right)}_{\text{alternative option value}} \tag{1}$$

where V_i^{ag}, V_i^{energy}, and V_i^{alt} represent the present values of agricultural production, energy development, and the best alternative use, respectively. The terms π_i^{energy} and π_i^{alt} denote the perceived probabilities that each option is realized. The energy option value is positive whenever

$V_i^{energy} > V_i^{ag}$ [6].

When a project enters the interconnection queue, it becomes public information and increases π_i^{energy} from zero to a positive value, capitalizing the energy development option into surrounding land prices. This prediction is consistent with empirical evidence showing that proximity to renewable energy development raises farmland values (Haan and Simmler, 2018; Abashidze and Taylor, 2023; Kunwar, 2024).

3.2 Project Withdrawal, Belief Revision, and the Role of Outside Options

When a project is withdrawn from the interconnection queue, it reduces the perceived probability of energy development. If $\pi_i^{energy,pre}$ denote the probability prior to withdrawal and $\pi_i^{energy,post} \approx 0$ after withdrawal, then the resulting price change can be expressed as:

$$\Delta P_i^{withdrawal} = -\pi_i^{energy,pre} \max\{V_i^{energy} - V_i^{ag}, 0\} + \Delta \left[\pi_i^{alt} \max\{V_i^{alt} - V_i^{ag}, 0\}\right] \tag{2}$$

The first term in the equation 2 captures the loss of the energy option and is almost always negative. The second term captures changes in the value of alternative land use opportunities following the withdrawal event. This term is zero in markets where no alternative development demand exists, in which case the net price effect of withdrawal is always negative. However, if the withdrawal of solar projects reveals or restores access to high-yielding alternative land use, then the second term is positive. This mechanism is empirically relevant to urban counties, as we discuss below. The net effect of project failure therefore depends on local market conditions and the availability of outside options.

In predominantly agricultural counties, the relevant outside option coincides with agricultural use, such that $V_i^{alt} \approx V_i^{ag}$. In this case, the second term is negligible and the price effect is negative, reflecting the removal of the previously capitalized energy option.

In urban-adjacent counties, the outside option is more valuable. Farmland near growing population centers is subject to competing demand from residential and commercial development,

[6] In equation 1, we distinguish the energy development option from other alternatives for clarity, as it is the margin directly affected by the activity of the interconnection queue, although it is itself one of the possible non-agricultural uses.

implying $V_i^{alt} > V_i^{ag}$ and potentially $V_i^{alt} > V_i^{energy}$. In these markets, an active solar project in the interconnection queue signals that the alternative land use may be restricted by a long-term energy lease for twenty-five to thirty-five years (Fiechter et al., 2024). Project withdrawal removes this restriction, and allows expectation to shift towards high-yielding alternative uses. In such settings, the net price effect can be written as:

$$\Delta P_i^{withdrawal} = -\pi_i^{energy,pre}\left(V_i^{energy} - V_i^{ag}\right) + \pi_i^{energy,pre}\max\{V_i^{alt} - V_i^{energy}, 0\} \quad (3)$$

The sign of net effect in equation 3 depends on the relative magnitudes of V_i^{energy} and V_i^{alt}. When alternative development yields higher returns than solar energy use, the restoration of the outside option exceeds the loss of the energy option.

A second mechanism that may reinforce positive impact of project withdrawal in urban-adjacent counties is the disamenities generated by solar panels for residential and commercial buyers, including visual impacts and land use conflicts (Van der Horst, 2007; Gaur and Lang, 2023; Jarvis, 2025). Project failure removes these anticipated disamenities, generating an additional upward price adjustment. This channel is largely absent in purely agricultural areas, where the demand of land for commercial and residential development is limited.

These mechanisms imply that project withdrawal reduces land values in agricultural markets but may generate different price responses in urban-adjacent markets, depending on the relative value of competing land uses. We test these predictions by estimating the effect of failed solar capacity on farmland prices and by examining heterogeneity across metro and non-metro counties.

4 Empirical Framework

We estimate the effect of failed solar development on farmland prices using transaction-level data. Our primary estimating equation relates the log sale price per acre of parcel i in county c at time t to the cumulative capacity of withdrawn projects in the county:

$$\ln P_{ict} = \alpha + \beta_1 \text{FailedMW}_{ct} + \mathbf{X}'_{ict}\boldsymbol{\gamma} + \mu_c + \lambda_t + \varepsilon_{ict} \quad (4)$$

where $\ln P_{ict}$ is the log of the sale price per acre for parcel i in county c at time t. FailedMW_{ct} is our variable of interest, and measures the cumulative capacity of solar projects that have been withdrawn from the interconnection queue in county c at year t. The vector $\mathbf{X}'_{ict}$ is the parcel-level controls including NCCPI soil quality, log acreage, and distance variables. μ_c represents county fixed effects that absorb all time-invariant differences across counties, and λ_t are year fixed effects.

Our coefficient of interest is β_1, which captures the effect of failed solar development on farmland prices. Our conceptual framework predicts $\beta_1 < 0$: as the project fails to materialize, the option value of future energy development which is embedded in land prices is eliminated, reducing the land price.

4.1 Identification Challenge

The estimation of equation 4 using Ordinary Least Square (OLS) regressions is unlikely to recover the causal effect of project withdrawal on farmland prices. The key identification challenge is that withdrawal decisions are not exogenous. First, developers select project locations based on a combination of observed and unobserved site characteristics. These locational features may include land availability, solar irradiance, transmission access, regulatory environment, and community acceptance, which may independently impact the farmland price dynamics. Second, projects have higher chance to fail in areas where permitting costs are highest, local opposition is strongest, or where expected returns to development are lowest. If any of these factors directly influence land price, OLS estimates of β_1 may reflect underlying differences in local fundamentals rather than the causal effect of project failure.

The direction of the OLS estimate is theoretically ambiguous. If projects tend to withdraw from areas with weaker local economies, where agricultural returns are also declining, then OLS would attribute pre-existing price decline to the withdrawal biasing $\hat{\beta}_1$ downward. However, if developers preferentially target high-value agricultural areas with strong price trends and these areas also see more withdrawals due to higher network cost or regulatory constraints, OLS would understate the negative price effect of failure. Regardless of direction, the endogeneity of withdrawal decisions means that OLS estimates cannot be given a causal interpretation, motivating our instrumental variable approach.

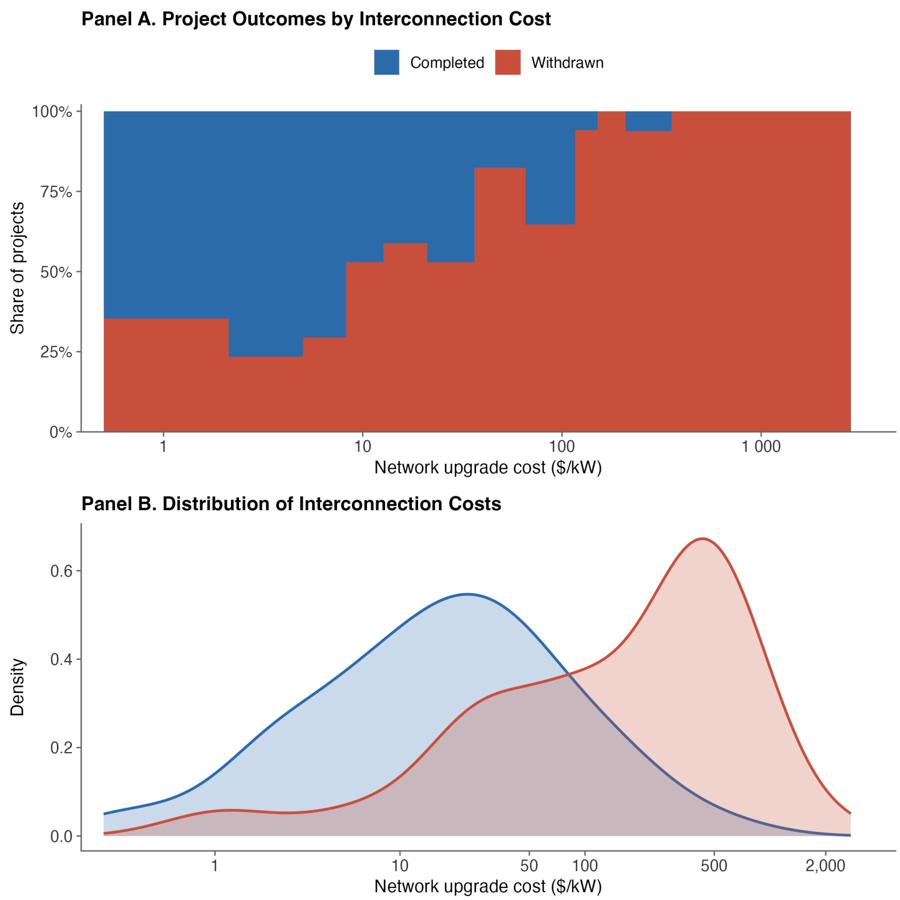
4.2 Instrumental Variable: Network Upgrade Costs

To address the endogeneity of project withdrawals, we exploit variation in network upgrade costs assigned during the interconnection study process as an instrument for failed solar capacity. These costs are a function of grid topology, line capacity, and system load at the proposed point of interconnection, all of which are physical characteristics of the existing transmission infrastructure. Because these factors are external to local land market conditions, they provide a plausibly exogenous source of variation in project viability.

The instrument satisfies two conditions required for valid identification. The first is *relevance*: network upgrade costs must be a strong predictor of project withdrawal. This condition is well-satisfied by the existing literature and by our first-stage results. According to Johnston et al. (2023), projects with second-study costs exceeding 0.1 million per megawatt are about 49 percentage points more likely to withdraw. This strong relationship holds even after accounting for permitting challenges and geography. Our first-stage results confirm a strong relationship in our sample, with an F-statistic of 262.7 well above conventional thresholds for instrument strength (Staiger and Stock, 1994; Stock and Yogo, 2002). Figure 2 illustrates this relationship directly, showing that projects with higher average network upgrade costs experience substantially higher rates of project withdrawal, consistent with the first-stage relationship underlying our identification strategy.

The second condition is the *exclusion restriction*: network upgrade costs must impact farmland prices only through its impact on project withdrawal, and not through any independent channel. We argue that this restriction is plausible for three reasons. First, interconnection costs are determined through engineering studies of the transmission network and reflect physical grid constraints rather than local land market conditions. While these costs may evolve with system congestion, they are primarily shaped by pre-existing infrastructures and do not directly affect

Figure 2: Interconnection Cost and Solar Project Outcomes



Notes: Panel A plots the share of completed and withdrawn solar projects within bins of network upgrade cost ($/kW). As costs rise, the withdrawal share increases monotonically, reaching nearly 100 percent at costs above $500/kW. Panel B plots the kernel density of network upgrade costs separately for completed and withdrawn projects. The distribution for completed projects is concentrated at lower cost levels, while the distribution for withdrawn projects is shifted substantially to the right. Together, these figures confirm that higher network upgrade costs are a strong predictor of project withdrawal, supporting the relevance condition of the instrumental variable strategy.

agricultural productivity, land quality, or other fundamental determinants of farmland prices.

Second, Johnston et al. (2023) show that observable project characteristics explain only about 41 percent of the variation in interconnection costs, and that costs are difficult to predict ex ante. This unpredictability reflects the fact that cost assignments depend on complex physical grid constraints, such as grid congestion, topology, and system load, that are not fully known to developers before the engineering study is complete. Because these factors are unrelated to local farmland fundamentals, the cost realizations provide an exogenous source of variation in project viability. This interpretation is further supported by the falsification test in Table A2, which shows that pre-queue farmland prices have no relationship with the network upgrade costs later assigned to projects in those counties.[7] Third, transmission infrastructure in the Midwest was largely developed to serve historical load patterns and centralized generation, rather than to accommodate future renewable energy siting (Davis et al., 2023; U.S. Department of Energy, 2023; Shreve et al., 2024). The cross-county variation in network upgrade costs therefore reflects historical grid architecture rather than contemporaneous economic conditions.

While the three arguments above establish the institutional plausibility of the exclusion restriction, we also provide supporting empirical evidence. A potential concern is that counties with high network upgrade costs may face systematically weaker grid infrastructure, which could independently affect farmland prices through channels unrelated to solar project failures. To assess this, we restrict farmland transactions occurring before any solar project entered the queue, and regress land prices with county's eventual network upgrade cost, controlling for parcel characteristics and year fixed effects. Under the exclusion restriction, network upgrade cost should have no predictive power for land prices before solar queue activity begins. Table A2 reports the results. The coefficient on network upgrade cost is economically negligible and statistically insignificant[8], indicating that counties facing higher engineering costs do not exhibit systematically different

[7]As an additional check, we add distance to the nearest transmission line in the baseline specification. The coefficient on Failed MW is unchanged to six decimal places, confirming that the substation distance control fully absorbs any independent effect of grid proximity on farmland prices.

[8]County fixed effects are excluded from this specification because the instrument, network upgrade cost, is measured at county-level. Including county fixed effects would therefore absorb most of the identifying variation in the instrument. As a check, we also estimate the falsification regression with county fixed effects; the coefficient on network upgrade cost remains statistically insignificant ($\hat{\beta}$ = 0.000, p = 0.762).

farmland price levels prior to any solar development activity. This test does not constitute proof of the exclusion restriction, which is fundamentally an untestable assumption. It does, however, provide direct empirical support that the cost variation reflects grid engineering constraints rather than underlying differences in local land markets.[9]

4.3 Two-Stage Least Squares Specification

We estimate the causal effect of failed solar development on farmland prices using a two-stage least squares (2SLS) estimator. In the first stage, we regress FailedMW_{ct} on our instrument network upgrade cost and all controls included in the second stage:

$$\text{FailedMW}_{ct} = \alpha_1 + \pi, \text{Z}_{ct} + \mathbf{X}'_{ict}\boldsymbol{\gamma}_1 + \mu_c + \lambda_t + \nu_{ict} \tag{5}$$

where Z is the network upgrade cost per kilowatt of the projects in county c at time t. π is the first-stage coefficient which captures the relationship between failed project capacity and engineering cost of the project.

Our second stage equation is:

$$\ln P_{ict} = \alpha_2 + \beta_1 \widehat{\text{FailedMW}}_{ct} + \mathbf{X}'_{ict}\boldsymbol{\gamma}_2 + \mu_c + \lambda_t + \varepsilon_{ict} \tag{6}$$

where $\widehat{\text{FailedMW}}$ is the predicted value of failed solar capacity obtained from the first stage equation 5. X_{ict} is a vector of parcel-level controls, and μ_c and λ_t represent county and year fixed effects, respectively.

5 Results and Discussion

We present our results in three steps. First, we show that network upgrade costs are the strong predictor of withdrawn solar capacity in the first stage of regression. Then, we estimate the causal

[9] Appendix Table A6 provides additional supporting evidence. Instrumenting failed wind capacity with the solar network upgrade cost yields an insignificant coefficient, consistent with the instrument operating specifically through solar project outcomes rather than through broader grid or rural economic conditions.

effect of failed solar development on farmland prices and show that the OLS estimates are biased towards zero, while instrumental variables reveal a negative and statistically significant price effect. Finally, we document substantial heterogeneity across the rural–urban landscape and assess whether the observed pattern is consistent with the outside-option mechanism developed in Section 3.2.

5.1 First-stage Estimates

Table 2 reports the first-stage estimates. This table shows that the network upgrade costs assigned during the interconnection study process are a strong predictor of the withdrawn solar capacity. A one dollar per kilowatt increase in average network upgrade cost in a county is associated with 0.20 additional megawatts of cumulative failed solar capacity. The coefficient remains stable across specifications as controls for realized and expected solar capacity are introduced, declining only modestly from 0.212 to 0.200, suggesting that the relationship between costs and failed projects is not driven by general solar activity in the county. This result is consistent with Johnston et al. (2023), who show that higher interconnection costs substantially increase the likelihood of project withdrawal. The corresponding first-stage F-statistics range from 262 to 361 across specifications, well above conventional thresholds for weak instruments (Staiger and Stock, 1994; Stock and Yogo, 2002), indicating that the instrument is strong.

5.2 Belief-Reversal

Table 3 reports OLS and 2SLS estimates of equation 4. OLS estimates (columns 1-3) of the effect of failed MW on land price are small, and statistically insignificant across all three specifications. Wu-Hausman tests[10] confirms that the endogeneity of project withdrawals bias the OLS estimates to zero and instrumental variables identification is necessary to estimate the true causal impact of project withdrawals. The direction of the bias is consistent with the positive selection, as explained in the section 4.1: developers target counties based on unobserved characteristics that

[10] The Wu–Hausman p-values range from 0.011 to 0.046, rejecting the null of exogeneity of failed MW across all specifications.

Table 2: First Stage Regression: Network Upgrade Cost as Instrument for Failed MW

Dependent Variable:	(1) Failed MW	(2) Failed MW	(3) Failed MW
Panel A: Instrument			
Interconnection Cost ($/kW)	0.212***	0.232***	0.199***
	(0.012)	(0.012)	(0.012)
Panel B: Solar Queue Controls			
Realized MW		0.147***	0.145***
		(0.005)	(0.005)
Expected MW			0.024***
			(0.001)
Panel C: Parcel Controls			
log(Acres)	−0.563**	−0.564**	−0.599***
	(0.229)	(0.227)	(0.227)
NCCPI	0.038***	0.040***	0.041***
	(0.013)	(0.012)	(0.012)
Panel D: Distance Controls (miles)			
Distance to Substation	−0.195	−0.225	−0.237
	(0.169)	(0.167)	(0.167)
Distance to Solar Farm	−0.173***	−0.135***	−0.123***
	(0.033)	(0.033)	(0.033)
County FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Observations	68,058	68,058	68,058
R^2	0.5295	0.5362	0.5382
F-stat (instrument)	298.11	359.98	262.16

Notes: This table reports first-stage estimates of the effect of network upgrade costs on Failed MW. The dependent variable is Failed MW, defined as the cumulative megawatts of solar projects withdrawn from the MISO Generator Interconnection Queue in county c through year t. Column (1) excludes Realized and Expected MW to show the reduced-form relationship between upgrade costs and failed projects. The results confirm that higher network upgrade costs strongly predict greater project withdrawals, with F-statistics far exceeding conventional weak instrument thresholds. Standard errors in parentheses. County and year fixed effects are included in all specifications.
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

independently support higher land values, partially offsetting the negative causal effect of project failure in simple OLS comparisons.

Columns 4–6 in table 3 reports our 2SLS estimates. Coefficients for Failed MW in 2SLS estimates are negative and statistically significant across all specifications. The preferred estimate[11] in column (4) indicates that each additional megawatt of failed solar capacity reduces farmland prices by approximately 0.37 percent. At the sample mean of 18.6 MWs of failed capacity per county-year, this implies a land price reduction of about 6.8 percent, or roughly $1,389 per acre at the sample mean price of $20,434.

This magnitude is broadly consistent with the potential value of the energy development option. Solar lease rates in the Midwest are typically three to five times prevailing cropland cash rents (Maguire et al., 2024; Winikoff and Parker, 2024; Fiechter et al., 2024). Given average cash rents of roughly $225 per acre in Indiana and Illinois, and only a fraction of the projects that are in queue are finally commercially operable, the expected premium associated with solar development can be substantial. A simple back-of-the-envelope calculation suggests that the resulting option value is of the same order of magnitude as our estimate.[12] Consequently, the estimated decline of approximately $1,389 per acre appears economically plausible and consistent with the loss of a partially capitalized energy development option.

These results provide direct evidence of the belief-revision mechanism central to asset pricing theory in farmland markets. By asset pricing theory, the option value of energy development projects is capitalized into farmland prices based on expectations of the use of the land in the future (Haan and Simmler, 2018; Plantinga et al., 2002; Abashidze and Taylor, 2023; Kunwar, 2024). Our results point to the other side of this relationship. When development expectations are reversed

[11]Column (4) is our preferred specification because it conditions on parcel-level controls and fixed effects without conditioning on other solar development outcomes. Columns (5) and (6) add Realized MW and Expected MW to assess whether the estimated effect of Failed MW is sensitive to controlling for completed and active solar development in the county. Because these variables are closely related to the same interconnection process that determines project failure, we interpret these specifications as sensitivity checks rather than preferred estimates. The coefficient on Failed MW remains stable across columns (4)–(6), indicating that the main result is not sensitive to this modeling choice.

[12]As a back-of-the-envelope calculation, suppose cropland cash rent is $225 per acre and solar lease payments are three to five times larger. The annual premium from solar conversion is then approximately $450-$900 per acre. If 25–30 percent of queued projects ultimately reach commercial operation (Rand et al., 2024), the probability-weighted annual premium is roughly $113-$270 per acre. Discounting this stream over a 25-30 year lease horizon at discount rates between 5 and 7 percent implies a present value of approximately $1,300-$4,200 per acre. The estimate of roughly $1,389 per acre therefore lies within the lower portion of this plausible range, consistent with partial capitalization of the solar development option.

Table 3: OLS and 2SLS Estimates: Effect of Solar Development on Farmland Prices

	OLS			2SLS		
Dependent Variable: Log(Price/Acre)	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Key Variables						
Failed MW	0.000009	−0.000029	−0.000011	−0.003752**	−0.003209**	−0.003213**
	(0.000098)	(0.000098)	(0.000098)	(0.001492)	(0.001363)	(0.001600)
Realized MW		0.000409***	0.000412***		0.000861***	0.000861***
		(0.000121)	(0.000121)		(0.000229)	(0.000255)
Expected MW			−0.000086**			0.000001
			(0.000035)			(0.000056)
Panel B: Controls						
log(Acres)	−0.590***	−0.590***	−0.590***	−0.592***	−0.592***	−0.592***
	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)
NCCPI	0.0113***	0.0113***	0.0113***	0.0114***	0.0114***	0.0114***
	(0.0003)	(0.0003)	(0.0003)	(0.0003)	(0.0003)	(0.0003)
Distance to Substation (mi)	−0.0126***	−0.0127***	−0.0127***	−0.0134***	−0.0135***	−0.0135***
	(0.0043)	(0.0043)	(0.0043)	(0.0044)	(0.0043)	(0.0043)
Distance to Solar Farm (mi)	−0.0052***	−0.0051***	−0.0051***	−0.0058***	−0.0055***	−0.0055***
	(0.0008)	(0.0008)	(0.0008)	(0.0009)	(0.0009)	(0.0009)
County FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	68,058	68,058	68,058	68,058	68,058	68,058
R^2	0.2194	0.2195	0.2196	0.2023	0.2074	0.2074
Panel C: IV Diagnostics						
Weak Instruments F				298.11***	359.98***	262.16***
Wu-Hausman p-value				0.011	0.018	0.043

Notes: This table reports OLS and 2SLS estimates of the effect of failed solar capacity on farmland prices. The dependent variable is the log of farmland sale price per acre. Columns (1)–(3) report OLS estimates and columns (4)–(6) report 2SLS estimates, in which Failed MW is instrumented using the network upgrade cost ($/kW). Each successive column adds Realized MW and Expected MW as controls. The results show that OLS estimates are small and statistically insignificant, while 2SLS estimates are negative and significant across all specifications. The preferred specification in column (4) implies that each additional megawatt of failed solar capacity reduces farmland prices by approximately 0.37 percent. The Wu-Hausman test rejects the null of exogeneity of Failed MW, confirming that IV identification is necessary.
Standard errors in parentheses.
County and year fixed effects are included in all specifications.
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

due to project failure, the option value previously capitalized into land prices is eliminated, resulting in a downward price adjustment. To our knowledge, our study is the first large-sample causal evidence of belief reversal in agricultural land markets.

Several additional checks support the main estimates. Figure A3 presents an annual weighted event study relative to the year of first solar project withdrawal. [13] Pre-withdrawal coefficients are positive, but statistically insignificant from zero, providing no evidence of differential price trends prior to project withdrawal. Following the first withdrawal, the coefficients become negative and statistically significant, consistent with the belief-revision mechanism. We interpret this evidence as complementary to the main IV estimates rather than as the primary source of identification. Appendix Table A1 shows that replacing the cumulative failed MW stock with an annual withdrawal flow measure yields a similar negative and statistically significant coefficient. Similarly, Table A5 confirms that the result is insensitive to outlier exclusion, whether trimming by failed capacity, farmland price, or both simultaneously. As a placebo check, we instrument failed wind capacity in the same counties using the solar network upgrade cost. The coefficient is small and statistically insignificant (Appendix Table A6), confirming that the estimated effect is specific to solar project outcomes and providing additional support for the exclusion restriction.

5.3 The Role of Outside Options

The conceptual framework in Section 3.2 predicts that the price effect of project failure depends on the availability of competing land use options. In markets where no valuable alternative uses exist, project withdrawal eliminates the energy development option and generates a negative price effect. However, in markets with alternative uses that yield higher returns than energy development and may be constrained by long-term solar leases, withdrawal can restore access to these options and partially or fully offset the loss. We test this prediction by estimating heterogeneous treatment effects across local land markets.

Table 4 reports the results from a 2SLS specification that allows the effect of Failed MW to differ

[13] The event-study is weighted by total number of transactions and acres transacted within each county-year. It is common practice to use weighted estimation when studying dynamic treatment effects in thin asset markets, to ensure that estimates are not driven by sparsely traded observations (see Case and Shiller, 1987; Amihud, 2002; Solon et al., 2015).

between metro and non-metro counties. We instrument both Failed MW and its interaction with metro status using network upgrade costs and their interaction with metro status. Both instruments are strong, with first-stage F-statistics of 173 and 407, respectively, and the Wu–Hausman test rejects the joint exogeneity of the endogenous regressors.The baseline coefficient on Failed MW captures the price effect in non-metro counties and is negative and highly significant. Each additional megawatt of failed solar capacity reduces farmland prices by approximately 4.2 percent. This large effect is consistent with the limited availability of alternative land-use opportunities in predominantly agricultural areas. In these counties, solar development represents one of the few opportunities to earn returns above traditional agricultural income, so project withdrawal removes a valuable development option and leads to a downward revision in land values. These estimates can be interpreted as the local average treatment effects for non-metro counties, where variation in project failure is driven by grid-engineering constraints rather than by economic conditions correlated with land values.

In metro counties, the price-effect looks different. The interaction coefficient is positive, nearly identical in magnitude to the baseline coefficient, and highly significant, implying that the overall price-effect in metro areas is close to zero. This finding is consistent with the outside option mechanisms discussed in Section 3.2. In areas where residential, commercial or industrial development is a viable alternative and yields higher returns than solar energy use, the failure of a solar project does not simply remove one option but can restore access to other, potentially more valuable land uses, offsetting the loss from energy development. In addition, anticipated solar development can generate disamenities for residential and commercial buyers, such as visual impacts and land use conflicts (Van der Horst, 2007; Gaur and Lang, 2023; Jarvis, 2025). Project withdrawals remove these concerns and provide additional upward pressure on land prices. This channel is largely absent in purely agricultural counties. Consistent with this interpretation, the split-sample estimates in Table A3[14] further show that the metro price effect is positive and statistically significant.

The metro-nonmetro classification provides a discrete measure of development opportunities, but urbanization varies continuously across space. We therefore estimate an alternative specifi-

[14]We prefer Table 4 given its stronger instruments and consistent complier population.

Table 4: Heterogeneous Effects of Solar Project Failure: Interaction IV Estimates

Dependent Variable: Log(Price/Acre)	(1) 2SLS
Panel A: Key Variables	
Failed MW	-0.0409^{***}
	(0.0092)
Failed MW × Metro	0.0407^{***}
	(0.0089)
Panel B: Controls	
log(Acres)	-0.601^{***}
	(0.010)
NCCPI	0.0111^{***}
	(0.0005)
Distance to Substation (mi)	-0.0187^{***}
	(0.0071)
Distance to Solar Farm (mi)	-0.0075^{***}
	(0.0015)
County FE	Yes
Year FE	Yes
Observations	68,058
Panel C: IV Diagnostics	
Weak Instruments (Failed MW)	171.14^{***}
Weak Instruments (Failed MW × Metro)	403.36^{***}
Wu-Hausman p-value	< 0.001

Notes: This table reports 2SLS estimates of the heterogeneous effect of solar project failure on farmland prices by metro status. Metro is an indicator equal to one if the county is classified as metropolitan by the USDA Economic Research Service (ERS) county typology. The results show that project failure significantly reduces farmland prices in non-metro counties, while the total effect in metro counties is near zero, reflecting the offsetting role of alternative land use options.
Standard errors in parentheses.
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

cation that allows the effect of project failure to vary with county population. Table 5 examines heterogeneity using county population. In this table, column (1) presents estimates based on a median population split while column (2) interacts Failed MW with log county population. Both of these specifications yield similar result. The negative effect of solar project failure is concentrated in lower-population counties, where alternative land-use opportunities are relatively limited. Figure 3 plots the continuous effect of this relationship using the specification that interacts Failed MW with population. The marginal effect of project failure remains negative throughout most of the population distribution, but eventually becomes positive in the most urbanized counties. Net marginal effects evaluated at selected population percentiles are reported in Table A4. This continuous gradient reinforces the theoretical prediction that the degree of outside-option availability, rather than a discrete rural-urban threshold, governs the direction and magnitude of the price response to project failure.

Table 5: Heterogeneity by County Population

Dependent Variable: Log(Price/Acre)	Median Split (1)	Log Population (2)
Panel A: Key Variables		
Failed MW	−0.053***	−0.251***
	(0.014)	(0.074)
Failed MW × High Population	0.052***	
	(0.013)	
Failed MW × log(Population)		0.020***
		(0.006)
High Population	0.024	
	(0.095)	
log(Population)		0.080**
		(0.043)
Panel B: Controls		
log(Acres)	−0.586***	−0.595***
	(0.011)	(0.012)
NCCPI	0.0126***	0.0123***
	(0.0007)	(0.0007)
Distance to Substation (mi)	−0.017**	−0.020**
	(0.008)	(0.009)
Distance to Solar Farm (mi)	−0.002	−0.005**
	(0.002)	(0.002)
County FE	Yes	Yes
Year FE	Yes	Yes
Observations	68,058	68,058

Notes: Both columns report 2SLS estimates instrumented by network upgrade cost ($/kW). In column (1), High Population is an indicator for counties above the median population (38,057). Net marginal effects evaluated at selected population percentiles are reported in Appendix Table A4.
Standard errors in parentheses.
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

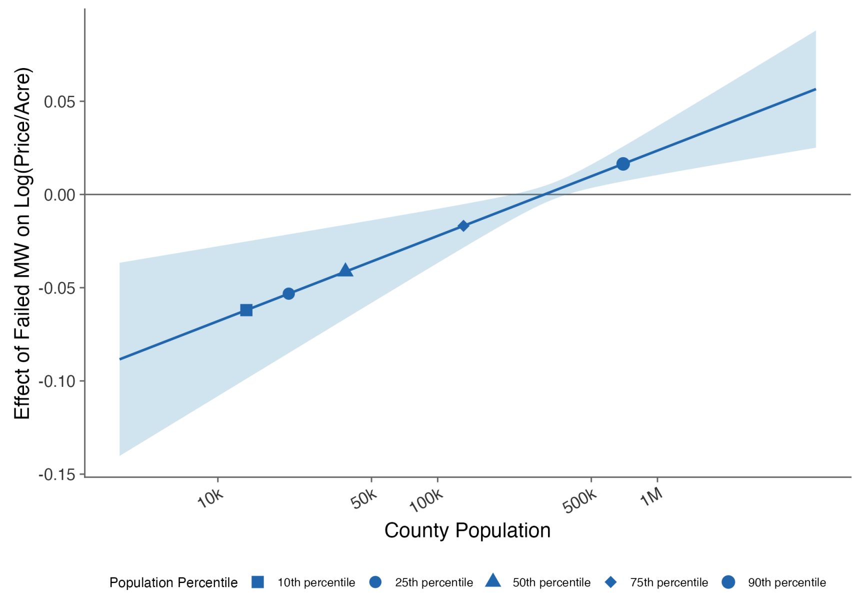


Figure 3: Effect of Project Failure on Farmland Prices by County Population

Notes: This figure plots the net marginal effect of Failed MW on farmland prices as a function of county population. The shaded band represents the 95 percent confidence interval. Labeled points correspond to the 10th, 25th, 50th, 75th, and 90th percentiles of the county population distribution. The effect is negative and significant across the lower four percentiles of the population distribution, and turns positive and significant at the 90th percentile, consistent with the restoration of alternative land use options in urban counties.

6 Conclusion

This paper examines how farmland prices respond when proposed solar energy projects fail to materialize. When a developer files an interconnection request, the project enters the public record and capitalizes an energy development option into surrounding land values. However, many of the queued projects are subsequently withdrawn, and with the updated withdrawal information, the capitalized expectation must be revised downward. Using approximately 68,000 parcel-level farmland transactions between 2016 and 2022, we estimate the causal effect of this belief-reversal mechanism on farmland prices. Because developers select project locations based on unobservables correlated with land values, OLS estimates are attenuated toward zero by positive selection bias. We address this endogeneity by instrumenting for failed solar capacity using network upgrade costs assigned during the interconnection study process. These costs are determined by engineering characteristics rather than local economic conditions, and the instrument is strong across all specifications, with first-stage F-statistics ranging from 263 to 407, and Wu-Hausman tests reject the null of exogeneity in every case.

We find that project failure reduces farmland prices on average, but the welfare cost of this failure falls almost entirely on rural communities. The preferred 2SLS specification implies a price decline of approximately 6.8 percent at the sample mean of failed capacity, or roughly $1,389 per acre, which is consistent with the partial capitalization of an energy development option. In non-metro counties, where energy development represents one of the few alternative land use options with returns higher than agricultural income, each additional megawatt of failed capacity reduces farmland prices by approximately 4.2 percent. In metro counties, the total effect is near zero, the loss of the energy development option is offset by the restored value of alternative residential and commercial uses. This rural-urban contrast holds continuously across the population distribution. The price effect of project failure is large and negative in low-population counties, diminishes monotonically as population increases, and turns positive at the 90th percentile of the county population distribution, consistent with the outside-option mechanism operating along a continuous gradient rather than a discrete threshold.

These findings have important policy implications, especially for energy, land markets, and rural communities. For interconnection and energy siting policy, option value capitalizes at the point of queue entry rather than construction, meaning each withdrawals imposes real economic losses on rural communities well before a single panel is installed. Policies that reduce project withdrawals may therefore generate welfare benefits for landowners that extend far beyond their direct impact on generation capacity. For landowners and farmers, the results reveal a bilateral risk that is rarely acknowledged in lease negotiations or land valuation practice. Farmland prices rise in anticipation of development when a project enters the queue, but if the project withdraws, prices fall, leaving landowners who purchased at inflated values exposed to a capital loss of approximately $1,389 per acre. For land policy makers, the findings provide an empirical basis for incorporating energy development expectations into farmland valuation frameworks. Current practice typically values agricultural land on agricultural fundamentals alone, but in counties with active interconnection queue filings, a meaningful share of observed transaction prices reflects energy option value that may or may not be realized.

There are several questions that remain open for future research. First, the mechanism in this paper operates through expectation formation, yet we know relatively little about how and when rural landowners become aware of queue filings or how quickly that information diffuses locally. Richer data on information diffusion would allow much sharper tests of when option value capitalizes upon queue entry. Second, we find a near-zero total effect in metro counties, and the option value channel, the disamenity channel, or both may be contributing to this result. The current design cannot separate them, and a structural model that exploits variation in the intensity of disamenity exposure would allow identification of each margin separately. Finally, this paper is restricted to solar projects in Indiana and Illinois, providing the first causal evidence on how farmland markets respond to failed energy development expectations. This setting, with its strong instruments, rich transaction data, and clear rural-urban variation, allows clean identification of the mechanism. Extending the analysis to wind energy and other states where agricultural and alternative land uses compete differently would test the external validity of these findings. Richer data on the full project lifecycle across energy types would open a broader research agenda into how development expectations form, evolve, and unravel in rural land markets across different technologies, regulatory environments, and regions of the country.

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Appendix

Table A1: Robustness: Annual Withdrawal Flow Measure

Dependent Variable: Log(Price/Acre)	2SLS
Panel A: Key Variable	
Withdrawn MW (annual flow)	-0.027^{**}
	(0.014)
Panel B: Controls	
log(Acres)	-0.596^{***}
	(0.008)
NCCPI	0.0119^{***}
	(0.0005)
Distance to Substation (mi)	-0.0173^{**}
	(0.0060)
Distance to Solar Farm (mi)	-0.0077^{***}
	(0.0017)
County FE	Yes
Year FE	Yes
Observations	68,058

Notes: Failed capacity is measured as MW withdrawn in year t only, rather than the cumulative stock used in the main specification. The negative and significant coefficient confirms the main result is not sensitive to the stock vs. flow construction.
Standard errors in parentheses.
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A2: Falsification Test: Network Upgrade Costs and Pre-Period Farmland Prices

Dependent Variable: Log(Price/Acre)	OLS
Network Upgrade Cost ($/kW)	-0.00001
	(0.001)
log(Acres)	-0.832^{***}
	(0.051)
NCCPI	0.0212^{***}
	(0.0031)
Distance to Substation (mi)	0.025
	(0.046)
Distance to Solar Farm (mi)	-0.022^{***}
	(0.008)
County FE	No
Year FE	Yes
Observations	610
R^2	0.341

Notes: Sample restricted to farmland transactions occurring before the first solar project entered the interconnection queue in the county. The insignificant coefficient supports the exclusion restriction: counties that will eventually face higher network upgrade costs do not exhibit systematically different farmland prices prior to any solar queue activity.
Standard errors in parentheses.
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A3: Heterogeneous Effects of Solar Project Failure on Farmland Prices

	Metro Counties		Nonmetro Counties	
Dependent Variable: Log(Price/Acre)	OLS	2SLS	OLS	2SLS
Panel A: Key Variable				
Failed MW	−0.000079	0.004863***	−0.000260**	−0.031575***
	(0.000163)	(0.001809)	(0.000127)	(0.006230)
Panel B: Controls				
log(Acres)	−0.677***	−0.677***	−0.505***	−0.512***
	(0.009)	(0.009)	(0.008)	(0.013)
NCCPI	0.0090***	0.0090***	0.0131***	0.0122***
	(0.0005)	(0.0005)	(0.0004)	(0.0008)
Distance to Substation (mi)	−0.0118*	−0.0112*	−0.0156***	−0.0216**
	(0.0064)	(0.0065)	(0.0057)	(0.0095)
Distance to Solar Farm (mi)	0.0002	−0.0006	−0.0035***	−0.0051***
	(0.0014)	(0.0015)	(0.0011)	(0.0018)
County FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	33,417	33,417	34,641	34,641
Panel C: IV Diagnostics				
Weak Instruments F		279.28***		39.62***
Wu-Hausman p-value		0.005		< 0.001

Notes: OLS and 2SLS estimates reported for metro and nonmetro subsamples separately. Failed MW is instrumented by network upgrade cost ($/kW) in the 2SLS columns. Metro classification follows the USDA ERS county typology. The positive and significant metro 2SLS coefficient is consistent with the restoration of alternative land use options following project withdrawal.
Standard errors in parentheses.
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A4: Net Effects of Failed MW by County Population Percentile

Population Percentile	Population	log(Population)	Net Effect
10th	13,479	9.509	-0.062^{***}
25th	21,014	9.953	-0.053^{***}
50th	38,057	10.547	-0.041^{***}
75th	131,086	11.784	-0.017^{***}
90th	698,046	13.456	0.016^{***}

Notes: Net marginal effects computed from column (2) of Table 5 as $\hat{\beta}_{\text{Failed MW}} + \hat{\beta}_{\text{Failed MW}\times\log(\text{Pop})} \times \log(\text{population})$. Significance tested via the delta method using `linearHypothesis()`. A positive net effect at the 90th percentile indicates that project failure raises farmland prices in the most urban counties, consistent with the restoration of alternative land use options. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A5: Robustness Checks: Outlier Trimming

	Dependent variable: log(Price/Acre)		
	(1) Trim Failed MW	(2) Trim Price	(3) Trim Both
Failed MW	-0.0038^{***}	-0.0043^{***}	-0.0041^{***}
	(0.0014)	(0.0013)	(0.0012)
log(Acres)	-0.5936^{***}	-0.4675^{***}	-0.4677^{***}
	(0.0060)	(0.0054)	(0.0054)
NCCPI	0.0113^{***}	0.0103^{***}	0.0102^{***}
	(0.0003)	(0.0003)	(0.0003)
Distance to substation (mi)	-0.0126^{***}	-0.0112^{***}	-0.0105^{***}
	(0.0044)	(0.0038)	(0.0038)
Distance to solar farm (mi)	-0.0051^{***}	-0.0044^{***}	-0.0041^{***}
	(0.0009)	(0.0008)	(0.0008)
Distance to transmission (mi)	-0.0256^{***}	-0.0157^{***}	-0.0157^{***}
	(0.0024)	(0.0021)	(0.0021)
County FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Observations	67,407	66,696	66,046

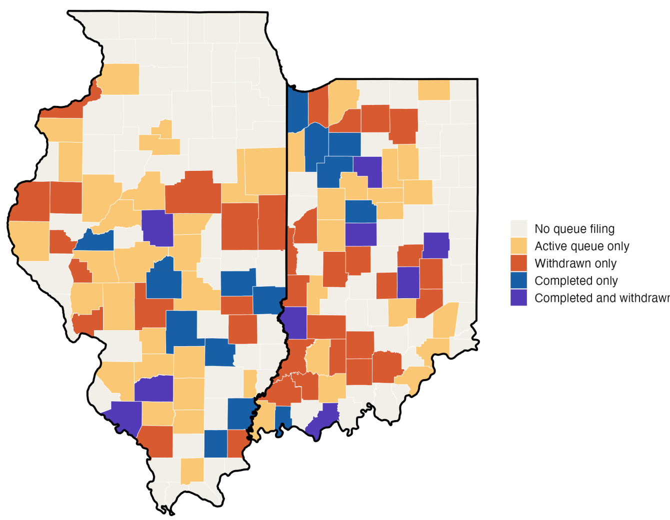
Notes: Column (1) excludes observations above the 99th percentile of accumulated failed capacity. Column (2) excludes parcels below the 1st and above the 99th percentile of price per acre. Column (3) applies both restrictions simultaneously.
Standard errors in parentheses.
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A6: Placebo Test: Failed Wind Capacity

	Dependent variable: log(Price/Acre)
Failed Wind MW	0.0077
	(0.0050)
log(Acres)	-0.5920^{***}
	(0.0102)
NCCPI	0.0115^{***}
	(0.0006)
Distance to substation (mi)	-0.0067
	(0.0084)
Distance to solar farm (mi)	-0.0043^{***}
	(0.0015)
Distance to transmission (mi)	-0.0266^{***}
	(0.0044)
County FE	Yes
Year FE	Yes
Observations	68,058

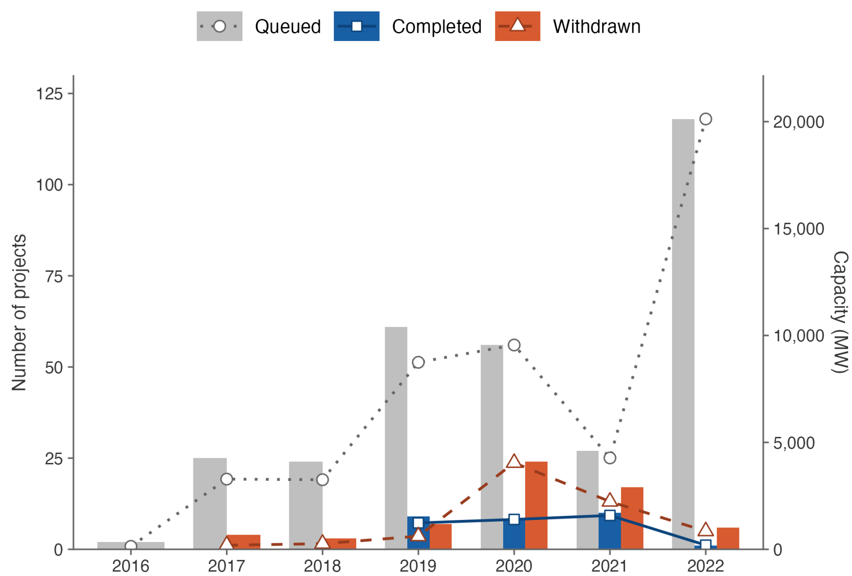
Notes: 2SLS estimates in which failed wind capacity is instrumented by the solar network upgrade cost ($/kW). The insignificant coefficient on Failed Wind MW confirms two things: the estimated effect is specific to solar project outcomes rather than reflecting a general energy development channel, and the instrument does not affect farmland prices through channels unrelated to solar project failure, providing additional support for the exclusion restriction. Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Figure A1: County-level solar project status based on MISO interconnection queue data



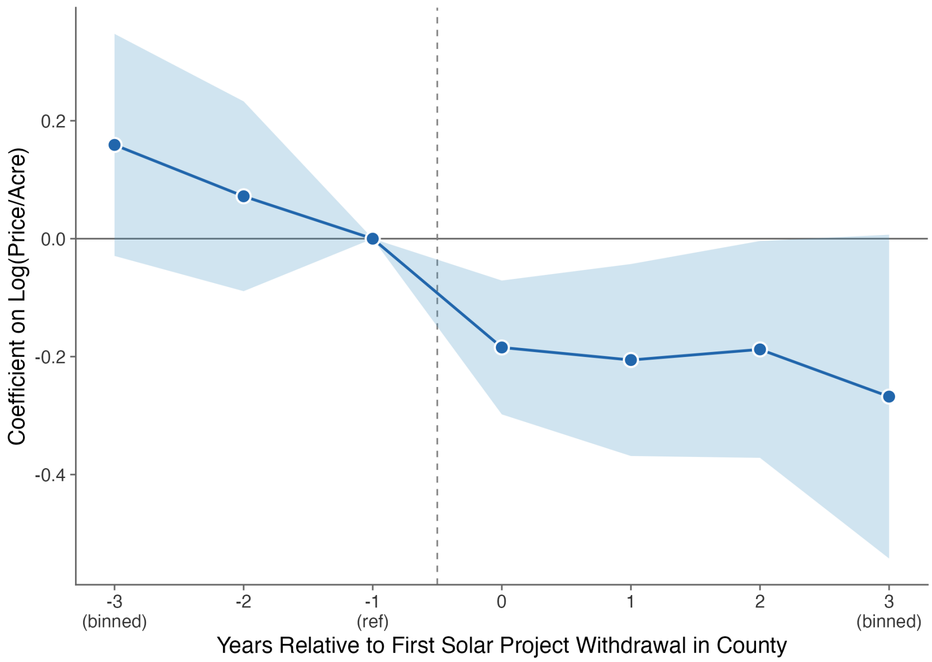
Notes: Each county in Indiana and Illinois is classified by the development status of solar projects in the MISO Generator Interconnection Queue as of the end of the sample period. Counties are categorized as having completed projects, withdrawn projects, active projects, or no projects. A county may appear in multiple categories if it contains projects at different stages of development.

Figure A2: Solar activity timeline



Notes: The figure plots the annual number of solar projects and solar capacity in Indiana and Illinois by development status. Data are obtained from the MISO Generator Interconnection Queue (GIQ). Sample period: 2016–2022.

Figure A3: Pre-trend



Notes: The figure plots event-study coefficients for the effect of solar project withdrawals on farmland price per acre, estimated relative to the year of first project withdrawal in a county. Event time zero corresponds to the year of first withdrawal; event time -1 is the omitted reference period. Coefficients in the pre-period are jointly insignificant. However, post-period coefficients are negative and statistically significant, supporting the belief-revision mechanism in which failed development expectations reduce the option value capitalized into farmland prices.